

		the force becomes negative.
10	B	When maximum friction force is experienced for each of the discs: <u>For m:</u> <i>friction, $f = mr\omega_1^2$</i>, where ω_1 is the angular velocity just as disc slips off $\omega_1^2 = \frac{f}{mr}$ <u>For 2m:</u> <i>friction, $(2f) = (2m)(2r)\omega_2^2$</i>, where ω_2 is the angular velocity just as disc slips off $\omega_2^2 = \frac{f}{2mr}$ As the angular velocity is gradually increased, it will reach ω_2 before ω_1. Hence the 2m disc will experience maximum friction
11	C	Applying conservation of linear momentum to the collision: